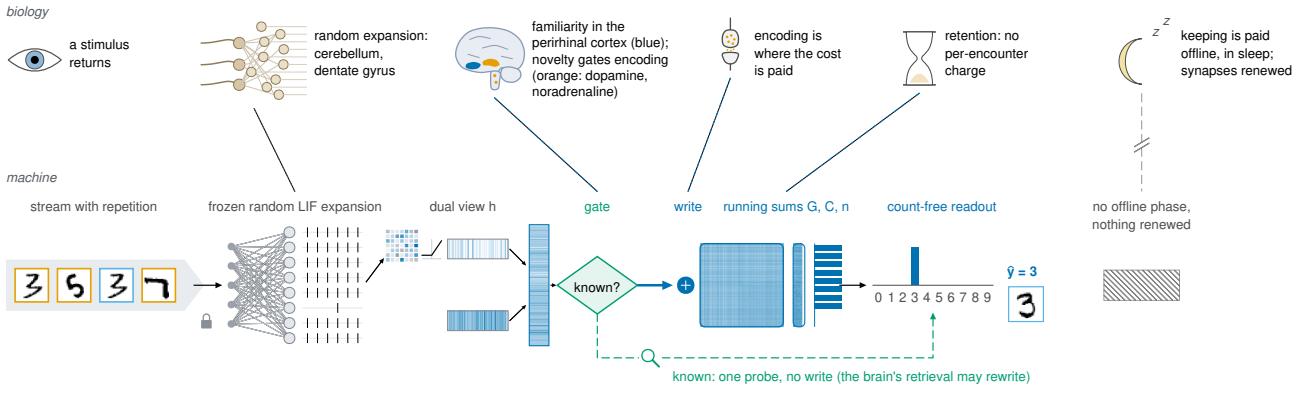
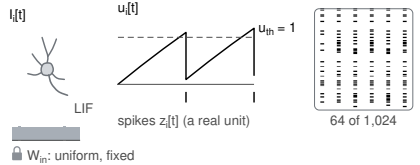


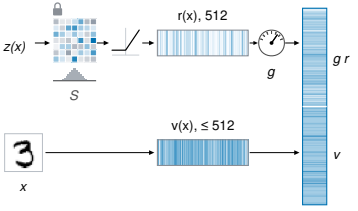
b Frozen spiking expansion



$u_i[t] = \beta\, u_i[t-1] + I_i[t]$, $\beta = 0.95$
$z_i[t] = 1$ if $u_i[t] > u_{th}$, else 0;
then $u_i[t] \leftarrow u_i[t] - u_{th}\, z_i[t]$
$z_i(x)$: the mean of $z_i[t]$ over $T = 6$ steps (20 on SHD)

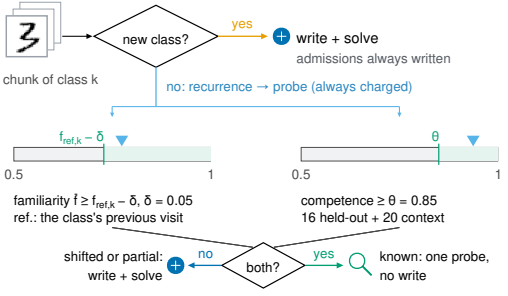
drive calibrated once to std 1.5, never trained;
lineage: the sparse random expansions of cerebellum and dentate gyrus

c Dual view



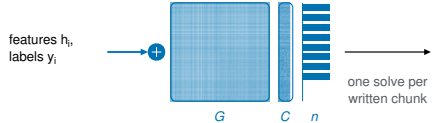
r(x) = ReLU(S^T z(x) + b), S Gaussian, fixed
h = [g r(x); v(x)], g = std(v) / std(r)
the view keeps what the spike counts discard

d Recognition gate (REWIRE-gated)



budgeted gate: a class is skipped only once B = 128 of its samples are written

e Write and solve



$G \leftarrow G + \Sigma\, h\, h^T$ (d × d)
$C \leftarrow C + \Sigma\, h\, e_y^T$ (d × K)
$n \leftarrow n + \Sigma\, e_y$ (K)

sums over the written samples:
the only plastic state;
order of writing does not matter

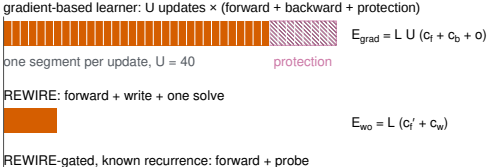
$\mu_k = C_k / n_k$, $S_w = G - \Sigma\, n_k\, \mu_k\, \mu_k^T$
$P = (S_w / (N - K) + \varepsilon\, \bar{\lambda}\, I)^{-1}$, $\varepsilon = 0.1$
$s_k(h) = h^T P\, \mu_k - ½\, \mu_k^T P\, \mu_k$
~~$+ \log(n_k / N)$~~ no such term: count-free
$\hat{y} = \text{argmax}_k\, s_k(h)$



order-invariant (Prop. 1) | count-free (Prop. 2)

memory $d^2 + dK + K$ numbers, whatever the stream's length; no stored samples, no backward pass

f What one chunk costs



bars schematic, not to scale; colours as in Figs 1 and 2